

ZeroDrift Mini-Rig — Build Guide (under ₹700)

NOT in the pitch deck, on purpose. The problem statement asks for validation *without* optical hardware, and our pitch is software-only. This rig is an optional table-side principle demo: a webcam finds a blinking phone flashlight and two tiny motors physically steer a laser onto it — the same closed loop our simulator runs, made touchable. If you build nothing, the ₹0 webcam demo ([docs/zero_cost_demo.md](#)) already works.

1. What you are building

Phone flashlight blinking at 4 Hz (the beacon) → **laptop webcam** sees it → our Python script identifies it *by its blink* (steady lights rejected) → sends angles to an **Arduino Nano** → two **SG90 servos** turn a **laser pointer** until its dot sits on/near the phone. Camera → detect → identify → steer → repeat, 15–30 times a second. That is coarse alignment, physically.

2. Shopping list (checked against Indian sellers, Aug 2026)

#	Item	Qty	Typical price	Where
1	SG90 9g micro servo	2	₹60–90 each	robu.in, robocraze.com, zbotic.in, local electronics market
2	Pan-tilt bracket kit for SG90 (2-axis)	1	₹80–170	robocraze.com "Servo Pan Tilt Setup", robu.in
3	Arduino Nano clone (CH340, soldered pins)	1	₹210–260	robu.in, hubtronics.in, dnatechindia.com (₹219), Amazon.in
4	KY-008 / 650 nm 5 V laser module (5 mW)	1	₹40–90	robu.in "Laser Module 650nm 5V"
5	Female-female jumper wires (20 pc)	1	₹60–90	any of the above
6	Mini-USB cable (Nano uses mini-USB!)	1	₹40–80	often lying at home — check first
—	Webcam	—	₹0 — laptop's built-in works	—
—	Beacon	—	₹0 — any phone + free "strobe light" app at ~4 Hz	—
	TOTAL		≈ ₹550–700	one robu.in / robocraze order ships it all

Tools: small Phillips screwdriver, tape/blu-tack, scissors. **No soldering.** Buying tips: order servos+bracket from ONE seller so the screws match; "CH340" in the Nano listing matters (that's the USB chip the driver needs).

3. Assembly — 30 minutes

1. **Bracket:** assemble the pan-tilt kit per its diagram: one servo sits flat (pan), the second mounts sideways on top (tilt). Don't fully tighten the servo-horn screws yet.
2. **Laser:** tape / blu-tack the KY-008 onto the top (tilt) platform, lens pointing forward.

3. **Wiring** (jumper wires, no soldering — servo plugs are 3-pin): Pan servo: orange→D9 red→5V brown→GND Tilt servo: orange→D10 red→5V brown→GND Laser KY-008: S→D7 middle→5V →→GND All three share the Nano's 5V and GND pins (use the bracket's wires or split with jumpers). Two SG90s on USB power is fine for this light load.
4. **Base**: tape the bracket to a heavy book so it can't tip.

4. Flash the Arduino — 10 minutes

1. Install Arduino IDE (arduino.cc). Windows: if the board isn't detected, install the **CH340 driver** (search "CH340 driver windows").
2. Open `tools/rig/rig_firmware.ino` from our repo.
3. Tools → Board: **Arduino Nano** · Processor: **ATmega328P (Old Bootloader)** (clone boards usually need Old Bootloader) · Port: the COM/tty that appeared.
4. Upload (→ arrow). Both servos snap to centre — that's success.

5. Run it — 5 minutes

```
pip install opencv-python numpy pyserial
python tools/rig/rig_track.py --port COM5          # Windows (see IDE for port)
python tools/rig/rig_track.py --port /dev/ttyUSB0  # Linux/Mac
```

1. Put the rig next to the laptop, both facing the room.
2. Second teammate stands 3–5 m away, phone strobing at ~4 Hz, facing the webcam.
3. Window shows **LOCKED** + green ring → laser turns on and walks onto the target.
4. **If the dot runs AWAY from the phone**: your servo orientation is mirrored — open `rig_track.py` and flip the sign of one gain (change `- args.gain*ex` to `+`, or the tilt line). One-character fix.
5. Keys: arrows = trim · c = centre · l = laser toggle · q = quit.

6. Calibration for the demo table

- Start with `--gain 0.02`. Wobbly/oscillating → lower it (0.01). Too slow → raise (0.04).
- Camera and laser should sit close together (parallax stays small).
- Demo in normal indoor light; a very bright window behind the target can out-shine the phone — face away from windows.

7. What to SAY while it runs

"Brightness is not identity — modulation is. The steady tube-light is brighter than the phone, but it doesn't blink at the agreed 4 Hz, so it scores zero and is rejected — same hard gate our simulator uses, where the tracker locked a wrong target 0 times in 64 randomised runs. And the loop you're watching — see, identify, steer — is the same loop our virtual terminal runs at 30 fps against a satellite."

8. Troubleshooting

Symptom	Fix
Nano not detected	Install CH340 driver; try another cable (data cable, not charge-only)
Upload error	Processor → ATmega328P (Old Bootloader)
Servos jitter constantly	Lower <code>--gain</code> ; check GND shared between Nano and servos
Never says LOCKED	Phone strobe closer to 4 Hz; move closer; dim room lights
Locks onto tube-light	It won't stay locked — score decays; if it does, raise threshold 0.35 → 0.5 in script
Laser dot far from phone	Normal small offset (parallax); mount laser closer to webcam

9. Video & reading references (real links, verified via search)

- YouTube — [OpenCV object tracking · Arduino laser pan/tilt](#)
- YouTube — [Face tracking with OpenCV, Python, Arduino & pan-tilt servos](#) (their loop is identical to ours — they centre a face, we centre a blinking beacon)
- Written — [PylImageSearch pan/tilt tracking tutorial](#)
- Written — [Top Tech Boy: pan/tilt servo object tracking in OpenCV](#)
- Code reference — [aju22/Object-Tracking-OpenCV_Arduino on GitHub](#)

10. Safety

5 mW red laser: never at eyes or faces; aim across a table at a paper target zone; firmware soft-limits (pan 20–160°, tilt 40–140°) keep it from sweeping the room — don't remove them.